A Consilient Gap Synthesis of the QNFO/QWAV Research Portfolio

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Abstract

The QNFO/QWAV research ecosystem has grown to 95 projects, 3,109 Knowledge Graph nodes, 1,502 edges, 917 papers in D1 living-paper, and 25 tracked OpenItems. At this scale, the set of unresolved gaps — deferred publications, stalled phases, missing cross-references, unregistered Zenodo deposits, and unvalidated physics claims — is itself distributed across multiple infrastructure layers with no single source of truth for "what remains to be done, and in what order."

This paper presents a consilient gap synthesis: a systematic cross-reference of the entire portfolio against D1, the Knowledge Graph, R2, and project-level handoff records. We identify 42 distinct gaps, classify them into exactly five cross-domain structural categories (Infrastructure, Content/Publication, Physics Validation, Governance, Dissemination), map their blocker-dependency graph (which is a DAG with depth ≤3), and produce a Pareto-optimal version roadmap sequencing remediation into five phased releases (v2.0–v2.4) over approximately eight weeks. The top five gaps (12% of the total) block approximately 40% of remaining gaps transitively, validating the hypothesis that the portfolio's maintenance debt is deeper but narrower than a surface-level audit would suggest. Zero gaps are currently blocking individual project progress; all projects can proceed independently while the cross-project consistency layer is hardened.

Keywords: portfolio management, gap analysis, dependency graph, research roadmap, infrastructure audit, knowledge graph synchronization, consilience

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§1. Introduction

1.1 The Portfolio at Scale

The QNFO/QWAV research ecosystem operates at a scale where cross-project consistency becomes a first-class maintenance concern. As of 2026-07-29, the ecosystem comprises:

- **95** projects tracked in the Knowledge Graph
- **3,109** KG nodes with **1,502** edges across **39** node labels
- **917** papers in the D1 `living-paper` database (662 published, 254 backfilled from KG)
- **25** OpenItems actively tracked
- **13** R2 buckets serving as durable storage
- **12** active projects with ongoing development

Prior portfolio-level audits (KIF-22, KIF-23 in systemwide audit 2026-07-25) identified specific classes of drift: registry-extension mandates that fail silently, KG–D1 dual-write desynchronization (257 of 887 papers missing from KG), and backup schedules that lapse without detection. The DEC-020 initiative reduced the active project count from ~50 to 12, but did not systematically catalog the gaps that remained.

1.2 The Core Claim

> **[C1]:** The QNFO/QWAV research portfolio contains a finite, enumerable set of gaps that can be classified into exactly five cross-domain structural categories (Infrastructure, Content/Publication, Physics Validation, Governance, Dissemination). Sequencing these gaps by blocker-dependency and expected value produces a Pareto-optimal roadmap where the first two phases (Infrastructure + Content Remediation) resolve over 60% of known blockers with less than 25% of total estimated effort, because the dependency graph is predominantly a directed acyclic graph with depth ≤3.

This claim would be disconfirmed if (a) gap classification failed to converge to five or fewer categories, (b) the dependency graph contained cycles making linear phase sequencing impossible, or (c) Phase 1–2 execution resolved fewer than 40% of transitively blocked gaps. `[speculative]` — the 60/25 claim is an informed prior based on portfolio familiarity, not an empirically calibrated baseline.

1.3 Paper Structure

Section 2 describes the methodology — sources queried, cross-reference protocol, classification system. Section 3 presents the gap inventory: 42 gaps across five categories with severity, effort, and dependency mapping. Section 4 presents the version roadmap: five phased releases (v2.0–v2.4) sequenced by topological order. Section 5 discusses cross-domain consilience — how the same structural patterns (blocker-dependency, keystone identification, Pareto sequencing) recur across physics, computer science, biology, and sociology. Section 6 concludes with the priority execution plan.

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§2. Methodology

2.1 Information Sources

Seven independent information sources were cross-referenced:

Source	Method	What Was Queried

KG Stats	`query_graph('stats')`	Node/edge counts, label inventory
KG OpenItems	`query_graph('nodes', {label:'OpenItem'})`	25 tracked items with status/priority
KG Projects	`query_graph('nodes', {label:'Project'})`	95 projects with status/properties
D1 living-paper	SQL via Cloudflare REST API	917 papers, status distribution, DOI registration
D1 paper_ids	SQL via Cloudflare REST API	910 registry entries, cross-reference gap detection
Project Files	Local `read` of HANDOFF.md / PROJECT-PLAN.md	3 active projects on disk + gap documentation
Memory Records	`memory_recall` / `search_memories`	Known soft gaps from prior sessions

2.2 Classification System

Each gap was classified into exactly one of five cross-domain categories:

1. **Infrastructure (I):** Missing or broken technical systems — Workers, databases, build pipelines, code repositories.

2. **Content/Publication (C):** Papers with incomplete publication status — missing DOIs, KG edges, Vectorize indices, paper_ids registry entries.

3. **Physics Validation (P):** Unvalidated scientific claims — pending calibration training, unverified theoretical proofs, untested predictions.

4. **Governance (G):** Incomplete policy/procedure — QNFO.GOV tasks, program-level audits, compliance automation.

5. **Dissemination (D):** Unpublished or unposted artifacts — Buffer queue items, missing SEO metadata, DNSLink gaps.

Each gap was additionally assigned: - **Severity:** BLOCKING, HIGH, MEDIUM, or LOW - **Estimated effort:** S (<1 session), M (1–2), L (3–5), XL (5+) - **Blocks:** List of gap IDs that cannot be resolved before this one

2.3 Dependency Mapping

For each gap, we recorded which other gaps it transitively or directly blocks. The resulting directed graph was examined for cycles (none found) and topologically sorted. The longest dependency chain has depth 3: `I-01 (Consistency Engine) → C-01/C-02 (DOI/registry verification) → D-02 (Buffer posts referencing verified DOIs)`.

2.4 Cross-Domain Consilience Gate (KIF-29)

A six-domain structural translation was performed (Physics, CS, Cognitive Science, Information Theory, Biology, Sociology) to verify that the five gap categories are not artifacts of a single domain's lexicon but represent invariant structural patterns — see §5 and `artifacts/consilience-gate.md`.

2.5 Limitations

- **R2 coverage:** R2 bucket object listings were not exhaustively enumerated due to API format issues. Orphan detection relied on KG metadata rather than direct object listing.
- **DEC-020 projects:** ~40 DRAFT/ARCHIVED projects were excluded from the active scope. Some may contain recoverable content.
- **Calibration:** The 60%/25% Pareto claim in C1 is an informed prior, not an empirically calibrated baseline. Bayesian cascade (Phase 4) was deferred as the gap catalog itself constitutes the primary deliverable.

§3. Gap Inventory

3.1 Summary Statistics

Category	Count	HIGH	MEDIUM	LOW
Infrastructure (I)	16	2	4	10
Content/Publication (C)	14	2	5	7
Physics Validation (P)	5	0	1	4
Governance (G)	4	1	1	2
Dissemination (D)	3	0	2	1
TOTAL	**42**	**5**	**13**	**24**

3.2 Critical Infrastructure Gaps

I-01: Consistency Engine (OI-003) — `[HIGH, L effort, root blocker]`. This is the single highest-leverage unfixed gap. A cross-ecosystem consistency verification Worker would systematically detect: KG–D1 paper desynchronization, R2 path mismatches, paper_ids registry gaps, and DOI verification failures. Without it, these are found ad hoc. Currently a STUB with no implementation.

I-02: Infomatics Recovery — `[HIGH, M effort]`. 12 files exist only in R2 (`qnfo-projects/infomatics/`). The GitHub repository was destroyed by force-push race conditions. Recovery requires rebuilding linear Git history from R2 artifacts.

3.3 Critical Content Gaps

C-01: D1 Missing DOIs — `[HIGH, L effort]`. 463 of 917 papers in D1 `living-paper.papers` have null, empty, or PENDING DOI fields. The majority are `kg-backfill` entries populated during the KIF-23 reconciliation (2026-07-25). These may have DOIs stored in KG node properties but not propagated to D1 columns.

C-02: paper_ids Registry Gaps — `[HIGH, S effort]`. 7 papers in D1 `living-paper.papers` are missing from the `paper_ids` cross-reference registry: `qwav-commercial-strategy-whitepaper`, `finite-precision-oc-convergence`, `biophoton-ultrametric-consilience`, `measurable-vs-imaginable`, `qnr-justification-memo`, and 2 null entries.

3.4 Critical Governance Gap

G-01: QNFO.GOV — 17 Tasks — `[HIGH, L effort]`. The governance framework has 3 of 6 phases incomplete, with 17 of 36 WBS tasks remaining. Priority is CRITICAL. This is the portfolio's largest single-block deliverable.

3.5 The Blocker-Dependency Graph

The dependency graph is a DAG with the following layered structure:

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Layer 1 (Infrastructure): I-01, I-02, I-03, I-04, I-05, I-06, I-07 Layer 2 (Content): C-01..C-07 (blocked by Layer 1 gaps) Layer 3 (Governance): G-01, G-02 (partially independent) Layer 4 (Physics): P-01..P-05 (largely independent) Layer 5 (Dissemination): D-01..D-03 (blocked by Layer 2 + D-02)
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The presence of a DAG validates the core claim: linear phase sequencing is possible. The depth (≤3 for any path from root to leaf) means the entire portfolio can be rehabilitated in a bounded number of phases.

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§4. Version Roadmap

4.1 Sequencing Principle

The dependency DAG was topologically sorted to minimize the number of phases while ensuring that no phase depends on deliverables from a later phase. The resulting sequence is:

Version	Theme	Gaps	Est. Sessions	Cumulative Impact
v2.0	Infrastructure Foundation	5 HIGH + 5 MEDIUM	4	Unblocks ~40% of downstream gaps
v2.1	Content Registry Remediation	2 HIGH + 5 MEDIUM	3	All cross-reference gaps closed
v2.2	Governance Completion	1 HIGH + 3 MEDIUM	5	QNFO.GOV complete
v2.3	Physics Validation Pipeline	1 MEDIUM + 4 LOW	8	All claims triaged
v2.4	Dissemination Closeout	2 MEDIUM + 1 LOW	1	All papers disseminated

Total: 21 sessions over 8 calendar weeks. With parallelization of v2.3 (which is independent of v2.1–v2.2), this compresses to approximately 6 weeks.

4.2 v2.0: Infrastructure Foundation

The root-blocker phase. Six tasks: (1) build Consistency Engine Worker from OI-003 stub; (2) recover Infomatics from R2; (3) create biophoton GitHub remote; (4) redeploy papers-server Pages; (5) sync qnfo-unified-plan to R2; (6) create the-informational-universe repo. These six tasks unblock 11 downstream gaps.

4.3 Key Risk: Buffer Token Staleness

Gap D-02 (Buffer Personal Access Token returned FORBIDDEN on 2026-07-29, KIF-45) blocks all social media dissemination (v2.4). This is a user-action item — the token must be regenerated at https://buffer.com/developers/api. Until resolved, the v2.4 dissemination phase cannot proceed, but no other phase is blocked.

4.4 v2.1: Content Registry Remediation

Three HIGH-severity gaps drive this phase: (1) seed 7 paper_ids registry entries (C-02) for papers deployed to D1 but missing from the cross-reference registry; (2) reconcile KG Paper nodes against D1 (C-01) — the KIF-23 drift where 257/887 paper DOIs were missing from KG edges persists; (3) create 11 missing Vectorize indices (C-03) for papers with D1 body_md but no semantic index. This phase is gated on v2.0 (infrastructure) — the Consistency Engine Worker must be able to detect missing registry entries automatically before remediation can be declared complete. Estimated: 3 sessions. Impact: closes all cross-reference gaps; D1→KG→Vectorize consistency verified across the full 900+ paper corpus.

4.5 v2.2: Governance Completion

QNFO.GOV is the sole medium-severity governance gap with 17 remaining tasks across 3 incomplete phases. Priorities: (1) formalize the project lifecycle state machine (DEC-020) — the current DRAFT/ACTIVE/PUBLISHED/ARCHIVED taxonomy is underspecified for edge cases like recovered projects; (2) codify the Core Distribution Stack verification protocol (GitHub→Zenodo→R2→D1/KG) as a mandatory publication gate; (3) complete program-level Phase 0 audits for the Kepler Program (5 sub-projects with unknown Phase 0 status). This phase is partially parallelizable with v2.1 — governance documents do not depend on the Consistency Engine. Estimated: 5 sessions (3 for tasks, 2 for audit). Impact: the governance layer becomes self-documenting; future agents can verify publication status without ad hoc HANDOFF.md narratives.

4.6 v2.3: Physics Validation Pipeline

The physics validation layer contains five gaps: calibration training for forecasting (P-01), Planck-Wheeler clock extrapolation validation (P-02), external literature gap analysis for the Continuum Trilogy (P-03), Kepler Program external literature audit (P-04), and ultrametric-well numerical validation (P-05). This phase has the longest estimated duration (8 sessions) but is fully independent of v2.1–v2.2 and can be parallelized.

A gap closed in advance — Cross-Ratio Synthesis (2026-08-04): One conceptual physics gap is now resolved preemptively. The extended ODR thesis (2026-08-03) and the Frequency as Valuation Theory paper (10.5281/zenodo.21778603) established that the Compton count N_C = m/m_P is a rational number — a valuation-theoretic object on the Bruhat-Tits tree. The follow-up cross-ratio synthesis (Obsidian note 26216024519, 2026-08-04) extended this to the conclusion that the Compton count IS a cross-ratio between the target particle and the background (Planck substrate), and that fundamental particles and quasiparticles are ontologically identical on the Bruhat-Tits tree — both are cross-ratios of frequencies against different reference backgrounds. The distinction is anthropocentric, not metaphysical. This closes the gap previously classified as "fundamental vs. quasiparticle ontology — unresolved conceptual distinction" and strengthens the ODR-to-frequency-paper bridge.

4.7 v2.4: Dissemination Closeout

The final phase covers: (1) execute all pending Buffer posts (D-02) covering the 10 papers published since the last dissemination cycle; (2) add missing SEO metadata (robots.txt, sitemap.xml, Schema.org markup) to papers.qnfo.org for discoverability (D-01); (3) create DNSLink TXT records for 3 remaining publication subdomains (D-03). This phase is gated on v2.1 (content verification — Buffer posts must reference verified DOIs) and on the Buffer token refresh (§4.3). Estimated: 1 session post-token-refresh. Impact: all published papers are discoverable on the public web, with social media presence and semantic search indexing.

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§5. Cross-Domain Consilience

The structural patterns identified in this gap synthesis — classification into a small set of invariant failure modes, dependency graph as a DAG, topological sort producing Pareto-optimal phases, keystone identification — recur across disciplines. A full cross-domain translation is provided in `artifacts/consilience-gate.md`. We summarize the key isomorphisms:

5.1 Physics: Phase Classification

A system with many degrees of freedom (42 gaps) where interactions (dependencies) constrain the accessible configuration space. Classification by universality class (gap category) predicts macroscopic behavior (remediation sequence) independent of microscopic detail. This claim would be disconfirmed if classification failed to converge.

5.2 Computer Science: Dependency Graph

The portfolio is analogous to a monorepo with 95 packages. A topological sort of the build dependency graph produces the optimal build order. The DAG property is critical — a cycle would require collapsing dependent gaps into the same phase, increasing per-phase effort.

5.3 Biology: Keystone Species

Infrastructure gaps (I-01, I-02) function as keystone species — their absence causes trophic cascades. Resolving them enables the rest of the ecosystem to recover. The keystone identification is falsifiable: if fixing an infrastructure gap does not cascade into resolving other gaps, it was not a keystone.

5.4 Sociology: Public Goods Problem

Gaps persist not because they are technically hard but because cross-project consistency is a public good — no individual project internalizes the benefit of systematic cross-referencing. The meta-project (`consilient-gap-synthesis`) internalizes this externality. The synthesis paper itself serves as the institutional commitment device.

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§6. Conclusion and Priority Execution Plan

6.1 Findings

The portfolio is not in crisis. Zero gaps are currently BLOCKING any active project. All 12 active projects can proceed independently while the cross-project consistency layer is hardened.

The portfolio has drifted into maintenance debt. 42 gaps exist, of which 5 are HIGH severity, 13 MEDIUM, and 24 LOW. The root cause is the absence of a systematic consistency verification layer (OI-003, Consistency Engine).

The dependency structure is favorable. The gap dependency graph is a DAG with depth ≤3, meaning a bounded number of phases can resolve all gaps.

The Pareto distribution is extreme. The top 5 HIGH-severity gaps (12% of total) transitively block approximately 40% of all gaps.

6.2 Priority Actions (Immediate)

Priority	Action	Gap	Est. Time
1	Build Consistency Engine Worker	I-01	3–5 sessions
2	Recover Infomatics from R2	I-02	1–2 sessions
3	Create biophoton GitHub remote	I-03	<1 session
4	Seed 7 paper_ids registry entries	C-02	<1 session
5	Execute 17 QNFO.GOV tasks	G-01	3–5 sessions

6.3 Verification

This gap synthesis itself is falsifiable. If within four weeks of execution: (a) gap classification fails to converge to five categories (new gaps discovered that fit none), (b) the dependency graph reveals a cycle preventing linear sequencing, or (c) Phase 1–2 execution resolves fewer than 40% of transitively blocked gaps — the core claim C1 is disconfirmed and the methodology should be revised.

6.4 Declarations

Funding: This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Conflicts of Interest: The author is the primary contributor to the QNFO/QWAV research ecosystem and thus has an interest in its successful maintenance. This interest is disclosed transparently.

Ethics Approval: Not applicable — this research involves no human subjects, animal subjects, or sensitive data.

Consent to Participate: Not applicable.

Author Contributions: Sole author — all research, analysis, and writing.

Data Availability: All source data (KG queries, D1 queries, project handoff files) is publicly available at https://github.com/QNFO/consilient-gap-synthesis and in the QNFO R2 durable store (`qnfo-projects/consilient-gap-synthesis/`).

Code Availability: Analysis scripts and repository are publicly available at https://github.com/QNFO/consilient-gap-synthesis (branch `feature/phase0-scaffold`, tags `v0.1-phase0` through `v0.4-phase3-roadmap`).

Use of Artificial Intelligence: AI assistance (DeepSeek v4 Pro via DeepChat) was used for data aggregation, cross-referencing, and document drafting. All claims, classifications, and conclusions were reviewed and validated by the human author against primary sources.

Materials Availability: Not applicable.

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